

MoSE: Mixture of Slimmable Experts for Efficient and Adaptive Language Models

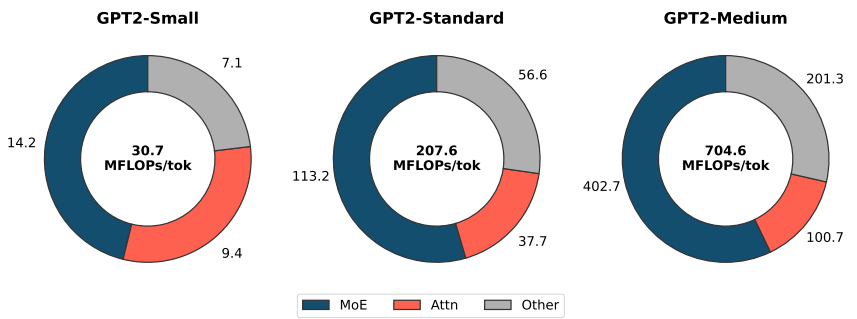
Motivation

MoE Leaves Capacity Rigid

- Sparse MoE routing activates only top-*k* experts, but each selected expert is still executed at **full width**.
- This makes accuracy-compute trade-offs coarse: a token either **pays for the whole expert** or **skips it entirely**.
- Experts dominate MoE compute**, so width is the natural control parameter.

Core Idea

MoSE decouples **expert selection** from **expert capacity**: the router picks experts, while an inference-time width policy decides how much of each active expert to use.



Method

MoSE: Recipe Card

Pre-Training / Continual Pre-Training

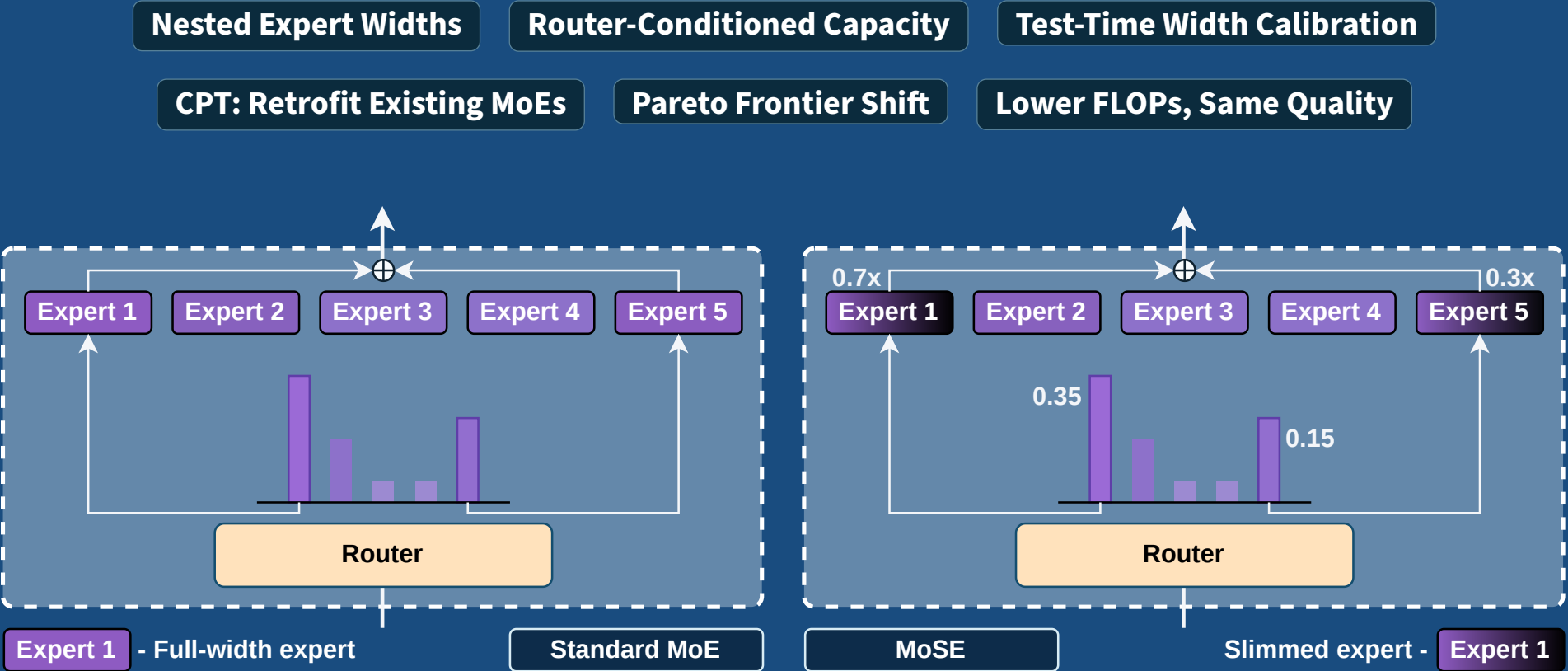
- Nested experts:** each MoE expert becomes $\mathcal{E}_i^w$; width $w \in \mathcal{A}$ activates hidden size $m(w)$ while reusing the same weights.
- One checkpoint:** train $w_{\max}$ plus one sampled $w \in [w_{\min}, w_{\max}]$ per batch; no separate model per budget.

Inference

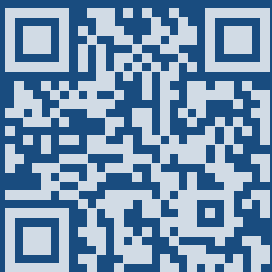
- Route as usual:** $\mathcal{G}(\mathbf{x})$ selects $\mathcal{S}(\mathbf{x})$; model, router, and expert weights stay frozen.
- Allocate capacity:** use $p_i(\mathbf{x})$ and total budget Γ to choose widths $w_i(\mathbf{x})$ for selected experts.
- Calibrate lightly:** TTT learns only γ , controlling how concentrated the width allocation is.

MoSE: Mixture of Slimmable Experts

MoE models choose **which experts** to run.
MoSE also chooses **how much of each expert** to execute.



A single pretrained checkpoint supports smoother accuracy-compute trade-offs by slicing active experts under a compute budget.

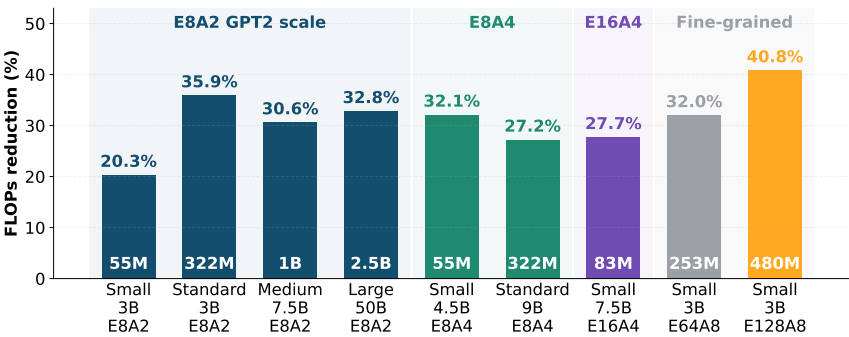


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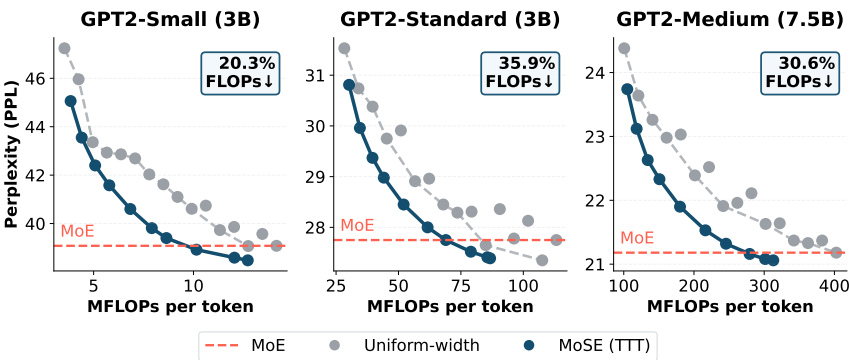
Experiments

Compute-Quality Frontier



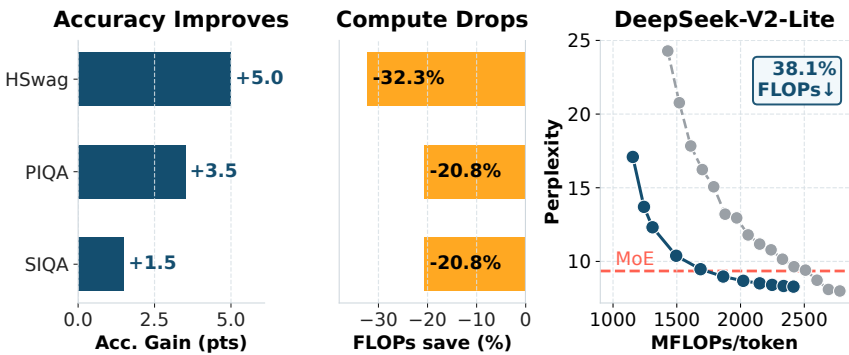
MoSE vs. MoE: Compute-quality frontier at matched perplexities.

Pareto Frontier Results



MoSE with test-time training traces a better Pareto frontier.

Zero-Shot Reasoning & Scaling to DeepSeek (CPT)



Zero-shot reasoning accuracy rises while FLOPs drop.
Scaling MoSE on DeepSeek using continual pre-training regime.

